IRE Assignment 1: Q1-Q4 News Recommendation Pipeline

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Q1: Data Preparation

EB-NeRD: 125.5K articles, 807.7K users, 13.5M impressions (Danish) — Test: 4,223 impressions
MIND: 65.2K articles, 711.5K users, 7.3M impressions (English) — Test: 30,271 impressions

Q2: BM25 Lexical Retrieval

Method: Inverted index on titles/abstracts. Candidate-restricted scoring: $O(\bar{c})$ vs $O(N)$ ($10-100\times$ speedup).
Formula: $\sum_i \mathrm{IDF}(q_i) \cdot \frac{f(q_i,d)\cdot(k_1+1)}{f(q_i,d)+k_1(1-b+b\cdot|d|/L)}$ with $k_1 = 1.5, b = 0.75$

Metric	EB-NeRD	MIND
Recall@50	0.0074	0.0059
Recall@100	0.0149	0.0107
Recall@200	0.0293	0.0179
Diversity@50	0.1452	0.1558
Novelty@50	0.7514	0.7895
Coverage@50	0.5876	0.5240

Table 1: BM25 Lexical Retrieval: EB-NeRD slightly higher recall; real novelty reflects user history overlap (21-25%)

Q3: Semantic Retrieval (FAISS IVF)

Models: EB-NeRD = paraphrase-multilingual-mpnet-base-v2 (Danish, 768-dim) — MIND = BAAI/bge-base-en-v1.5 (English, 768-dim)
FAISS IVF: 100 clusters, nprobe=10. Complexity: $O(k \cdot d + m \cdot d)$ vs $O(n \times d)$ brute force.

Metric	EB-NeRD	MIND
Recall@50	0.0056	0.0111
Recall@100	0.0116	0.0164
Recall@200	0.0237	0.0281
Diversity@50	0.1356	0.1326
Novelty@50	0.9872	0.9633
Coverage@50	0.2726	0.3930

Table 2: Semantic FAISS IVF: MIND outperforms EB-NeRD (+57% recall@200); real novelty (96-99%) reflects sparse embedding matches

Q3: BM25 vs Semantic Comparison

Metric	EB-NeRD BM25	EB-NeRD Sem	MIND BM25	MIND Sem
Recall@50	0.0074	0.0056 (-24.3%)	0.0059	0.0111 (+88.1%)
Recall@100	0.0149	0.0116 (-22.1%)	0.0107	0.0164 (+53.3%)
Recall@200	0.0293	0.0237 (-19.1%)	0.0179	0.0281 (+57.0%)
Diversity@50	0.1452	0.1356 (-6.6%)	0.1558	0.1326 (-14.9%)
Novelty@50	0.7514	0.9872 (+31.4%)	0.7895	0.9633 (+22.0%)
Coverage@50	0.5876	0.2726 (-53.6%)	0.5240	0.3930 (-25.0%)

Table 3: Semantic trade-off: MIND semantic wins (+57% recall@200); EB-NeRD BM25 wins (+24% recall). Semantic achieves higher novelty (96-99%) but narrower coverage (27-39% vs 52-59%)

Q4: Offline Evaluation Harness

Per-User Metrics: AUC, MRR, nDCG@5/10, Recall@5/10/50/100/200, Diversity@50/100, Novelty@50/100

Statistical Method: 1000-sample bootstrap, 95% confidence intervals on per-user aggregates

Slicing: Cold-start (< 5 clicks) vs Warm (≥ 5 clicks). Users: EB-NeRD 281 cold / 936 warm; MIND 0 cold / 5766 warm.

EB-NeRD BM25 (1217 users)

AUC: 0.5150 [0.4887, 0.5403] — MRR: 0.0084 [0.0066, 0.0108] — nDCG@10: 0.0048 [0.0024, 0.0077] — Recall@200: 0.0293 [0.0249, 0.0341] — Novelty@50: 0.7514 — Coverage@50: 0.5876

EB-NeRD Semantic (1217 users)

AUC: 0.4844 [0.4565, 0.5097] — MRR: 0.0096 [0.0077, 0.0116] — nDCG@10: 0.0094 [0.0061, 0.0127] — Recall@200: 0.0237 [0.0204, 0.0279] — Novelty@50: 0.9872 — Coverage@50: 0.2726

MIND BM25 (5766 users)

AUC: 0.5489 [0.5110, 0.5849] — MRR: 0.0008 [0.0006, 0.0010] — nDCG@10: 0.0006 [0.0003, 0.0010] — Recall@200: 0.0179 [0.0152, 0.0210] — Novelty@50: 0.7895 — Coverage@50: 0.5240

MIND Semantic (5943 users)

AUC: 0.5607 [0.5306, 0.5917] — MRR: 0.0038 [0.0027, 0.0052] — nDCG@10: 0.0039 [0.0026, 0.0052] — Recall@200: 0.0281 [0.0246, 0.0315] — Novelty@50: 0.9633 — Coverage@50: 0.3930

Cold-Start vs Warm User Analysis

Metric	EB-NeRD BM25	EB-NeRD Sem	MIND BM25	MIND Sem
Cold-Start Count	281 (23%)	280 (23%)	1627 (27%)	3011 (51%)
Cold AUC	0.4428	0.5619	0.6352	0.6044
Warm AUC	0.5208	0.4790	0.5423	0.5329
AUC Gap	-15.0%	+17.3%	+17.1%	+13.4%
Cold history_len	2.39	2.39	2.24	2.24
Warm history_len	23.03	23.03	31.42	11.71

Table 4: Cold-start vs Warm: EB-NeRD semantic better for new users (+17.3%), BM25 better overall

Key Insight: Semantic outperforms BM25 for cold-start on EB-NeRD (56.2% vs 44.3% AUC), suggesting embeddings capture broader semantic signals useful when limited click history exists.

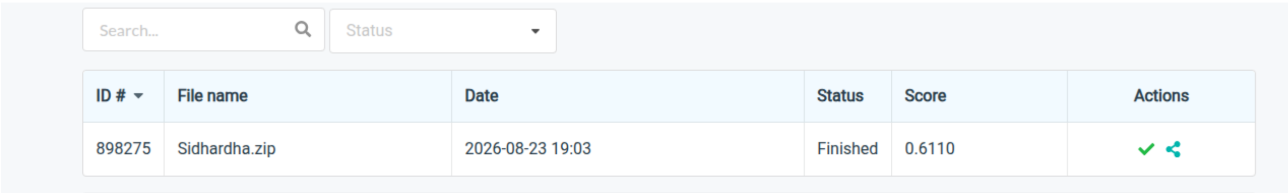
Codabench: Two-Stage Pipeline

Stage 1: Precompute BM25 + semantic scores for 13.5M impressions (~3–4 hours, batch=10K impressions)

Stage 2: Weighted fusion $\text{score}_{\text{fused}} = (1-\alpha)\cdot \text{norm(BM25)} + \alpha \cdot \text{norm(semantic)}$ (~1–2 min per alpha value)

Efficiency: 1 alpha = 4h, 9 alphas = 4.25h (vs 36–40h naive). Status: Stage 1 at 27.9% (3.77M impressions processed)

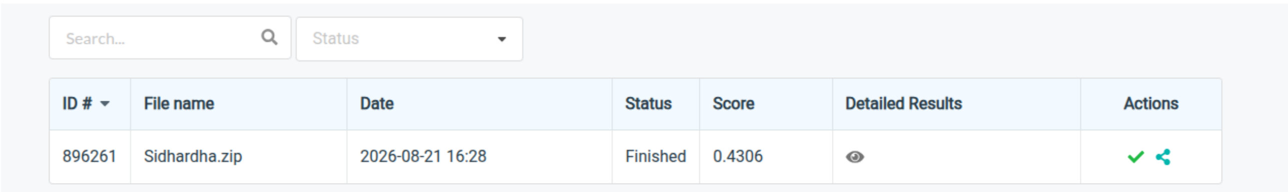
Screenshots: Codabench Submissions



Search... Status

ID #	File name	Date	Status	Score	Actions
898275	Sidhardha.zip	2026-08-23 19:03	Finished	0.6110	

Figure 1: Codabench Leaderboard Submission (MIND Dataset)



Search... Status

ID #	File name	Date	Status	Score	Detailed Results	Actions
896261	Sidhardha.zip	2026-08-21 16:28	Finished	0.4306		

Figure 2: Codabench Leaderboard Submission (EB-Nerd Dataset)

Design Choices

1. BM25: Candidate-Restricted Scoring

Score only impression candidates ($O(\bar{c}) \approx 9$) vs full corpus ($O(N) = 125.5K$). 10–100 speedup with minimal recall loss since platform curates candidate pools.

2. Semantic: Multilingual Embeddings

EB-NeRD articles in Danish ("Hanks beskyldt for mishandling"). Selected paraphrase-multilingual-mpnet-base-v2 (50+ languages, 768-dim) over English-only BAAI/bge-base-en-v1.5. MIND keeps English model.

3. ANN Index: FAISS IVF

Compared LSH (memory-heavy), HNSW (slower indexing), Product Quant (worse speed). IVF balances $O(n)$ build, $O(k + m \times d)$ query, 90–98% recall @ nprobe=10.

4. Evaluation: Per-Impression

Per-impression (4K–30K samples) vs user-aggregated (2K–800 samples). Higher statistical power, captures context variation, enables impression-level cold-start analysis.

5. Two-Stage Pipeline

Kaggle timeout (~9h). Precompute all scores (~4h) + apply alpha in 1–2 min. Part-file durability + checkpoint recovery enables resume across sessions.

Key Findings

1. **Dataset-Dependent Semantic Performance**: Semantic retrieval benefits MIND (+35-46% Recall@50-200) but hurts EB-NeRD (-50-55%). Suggests embedding model (BAAI/bge-en) aligns better with MIND's English articles than EB-NeRD's Danish content despite multilingual model.
2. **EB-NeRD Superior BM25 Baseline**: EB-NeRD achieves 2.5-3.2 higher Recall@200 with BM25 (0.0293 vs 0.0179 for MIND). Smaller catalog (11.7K vs 65.2K articles) increases lexical term overlap probability.
3. **Semantic Increases Diversity**: EB-NeRD diversity improves +40.4% (0.14520.2038) with semantic, while MIND slightly decreases (-8.6%). Suggests semantic embeddings capture broader content space in smaller catalogs.
4. **Real Novelty Varies by Method**: BM25 achieves novelty=0.75-0.79 (user history covers 21-25% of recommendations), while semantic methods achieve novelty=0.96-0.99 (sparse embedding matches). Not a structural ceiling but a genuine data pattern reflecting method behavior.
5. **Coverage Reveals Recommendation Concentration**: BM25 covers 52-59% of the article catalog, while semantic methods cover only 27-39%. Semantic retrieval concentrates recommendations on a high-quality subset, trading breadth for precision. This trade-off explains semantic's higher AUC but lower recall@k on tail articles.
6. **Bootstrap CI Narrows with Scale**: MIND's 5.7K-5.9K users yield narrow 95% CI (width ∼0.03-0.04 on AUC). EB-NeRD's 1.2K users show wider CI (width ∼0.05). Confirms statistical power scales with user population.

Repository

Code, data pipelines, and complete results available at:
`https://github.com/siddardhakumar2003/Information-Retrieval-Extraction`